Series and Convergence

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1 Series as limits of partial sums

First, an example

Before any formalism, watch a series *converge* numerically. Take the geometric series with ratio 1/2:

$$\sum_{n=0}^{\infty} \left(\tfrac{1}{2}\right)^n \;=\; 1 + \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{16} + \cdots \;=\; 2.$$

Read aloud: "the sum from n equals zero to infinity of one-half to the n equals two." The partial sums march visibly toward 2:

$$s_0 = 1, \quad s_1 = 1.5, \quad s_2 = 1.75, \quad s_3 = 1.875, \quad s_4 = 1.9375, \; \ldots$$

Read aloud: "s-zero equals one, s-one equals one point five," and so on. Each step halves the gap to the limit.

Contrast this with the *harmonic series* $\sum_{n=1}^{\infty} 1/n = 1 + 1/2 + 1/3 + 1/4 + \cdots$, whose terms also shrink to 0, yet whose partial sums grow without bound: $s_n \sim \ln n \to \infty$. Vanishing terms are *necessary* but not *sufficient* for convergence. The rest of this lesson explains exactly which series fall on each side of that line.

Given a real sequence $(a_n)_{n\geq 0}$, the symbol $\sum_{n=0}^{\infty} a_n$ denotes the formal infinite sum. Its value, when defined, is the limit of the sequence of partial sums.

Definition 1 (Series and partial sums)**.** The *n-th partial sum* of (a_n) is

$$s_n \;=\; \sum_{k=0}^{n} a_k.$$

Read aloud: "s-sub-n equals the sum from k equals zero to n of a-sub-k." The *series* $\sum_{n=0}^{\infty} a_n$ is the symbolic object associated with (a_n); we say it *converges to* $S \in \mathbb{R}$ iff $s_n \to S$ as $n \to \infty$, and write $\sum_{n=0}^{\infty} a_n = S$. Otherwise the series *diverges*.

Definition 2 (Absolute and conditional convergence)**.** The series $\sum a_n$ *converges absolutely* iff the series of non-negative terms $\sum |a_n|$ converges. If $\sum a_n$ converges but $\sum |a_n|$ diverges, the convergence is *conditional*.

It is a standard fact that absolute convergence implies convergence: the partial sums of $\sum a_n$ form a Cauchy sequence, dominated by the Cauchy tails of $\sum |a_n|$. The converse fails, as witnessed by $\sum (-1)^{n+1}/n = \ln 2$ paired with the divergent harmonic series.

Example 3 (Geometric series)**.** For $r \in \mathbb{R}$ with $|r| < 1$, $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$. Indeed $s_n = (1 - r^{n+1})/(1-r) \to 1/(1-r)$ as $r^{n+1} \to 0$. This is the workhorse comparator for the ratio test below.

2 The ratio test

The ratio test reduces convergence of $\sum a_n$ to comparison with a geometric series.

Theorem 4 (Ratio test)**.** *Let (a_n) be a sequence of nonzero real numbers and suppose*

$$L \;=\; \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$$

Read aloud: *"L equals the limit, as n goes to infinity, of the absolute value of a_{n+1} over a_n." exists in $[0,+\infty]$. Then:*

1. *if $L < 1$, the series $\sum_{n=0}^{\infty} a_n$ converges absolutely;*
2. *if $L > 1$ (including $L = +\infty$), the series diverges;*
3. *if $L = 1$, the test is inconclusive.*

Proof. We treat the three cases.

Case $L < 1$**.** Choose r with $L < r < 1$. Since $|a_{n+1}/a_n| \to L < r$, there exists $N \in \mathbb{N}$ such that

$$\left|\frac{a_{n+1}}{a_n}\right| \;<\; r \qquad \text{for all } n \geq N.$$

Read aloud: "the absolute value of a_{n+1} over a_n is less than r, for all n greater than or equal to N." Multiplying these inequalities for $n = N, N+1, \ldots, N+k-1$ gives

$$|a_{N+k}| \;<\; r^k\,|a_N| \qquad \text{for every } k \geq 0.$$

Read aloud: "the absolute value of a_{N+k} is less than r to the k times the absolute value of a_N, for every k at least zero." Therefore, for all $n \geq N$,

$$\sum_{m=N}^{n} |a_m| \;\leq\; |a_N| \sum_{k=0}^{n-N} r^k \;\leq\; \frac{|a_N|}{1-r}.$$

Read aloud: "the sum from $m = N$ to n of $|a_m|$ is at most $|a_N|$ times the sum from $k = 0$ to $n-N$ of r^k, which is at most $|a_N|$ over $1-r$." The partial sums of the non-negative series $\sum |a_m|$ are bounded above, hence converge (a monotone bounded sequence has a limit). Adding back the finitely many terms for $m < N$ shows $\sum |a_n| < \infty$, so $\sum a_n$ converges absolutely.

Case $L > 1$**.** Choose r with $1 < r < L$ (or $r > 1$ arbitrary if $L = +\infty$). There exists N such that $|a_{n+1}/a_n| > r$ for all $n \geq N$, hence $|a_{N+k}| > r^k|a_N|$. Since $r > 1$, $|a_n| \to \infty$, so $a_n \not\to 0$ and the series diverges by the term test.

Case $L = 1$**.** The series $\sum 1/n$ has $L = 1$ and diverges, while $\sum 1/n^2$ has $L = 1$ and converges. No conclusion can be drawn from L alone. $\square$

Example 5 (Exponential series)**.** For every $x \in \mathbb{R}$, setting $a_n = x^n/n!$ gives

$$\left|\frac{a_{n+1}}{a_n}\right| \;=\; \frac{|x|}{n+1} \;\longrightarrow\; 0.$$

Read aloud: "the absolute value of a_{n+1} over a_n equals absolute-x over $n+1$, which tends to zero." Thus $L = 0 < 1$, and Theorem 4 yields absolute convergence of $\sum_{n=0}^{\infty} x^n/n! = e^x$ for every real x. The same argument works for $x \in \mathbb{C}$ and, with $|x|$ replaced by $\|A\|$, for any square matrix A.

3 Power series

A *power series* centred at x_0 is a series of the form $\sum_{n=0}^{\infty} c_n(x - x_0)^n$. Its set of convergence is an interval (possibly a single point or all of $\mathbb{R}$) of *radius of convergence*

$$R = \frac{1}{\limsup_{n\to\infty} |c_n|^{1/n}} \qquad \text{(Cauchy–Hadamard).}$$

Read aloud: "R equals one over the limit superior, as n goes to infinity, of the n-th root of the absolute value of c_n — the Cauchy–Hadamard formula." Inside $|x - x_0| < R$ the series converges absolutely; outside it diverges; behaviour at the endpoints requires a separate test.

This perspective is the bridge from analysis to ML: every smooth activation function, the matrix exponential, and the leading-order expansion of the KL divergence are all instances of power-series approximation.